TAREK SHEREEN DARWISH

Software Engineer, Applied Machine Learning & AI

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SUMMARY

Applied ML engineer finishing a B.Sc. in Intelligent Systems Engineering. Built an explainable fraud-detection model (XGBoost, test ROC AUC 0.97, SHAP explanations written by Gemini), on-device TFLite glucose models for a wearable health band (Winner, Youth Leaders Foundation Hackathon 2026; 2nd place, Banha Hackathon), and the start of a local planner/executor LLM backend. Seeking a junior ML/AI engineering role, remote or relocation with visa sponsorship.

TECHNICAL SKILLS

ML & Data: scikit-learn, XGBoost, TensorFlow/TFLite, PyTorch, SHAP, NLP, computer vision, Pandas, Power BI, Tableau

LLMs: Gemini, LLaMA-70B, Gemma 3, local Qwen on llama.cpp, prompt engineering, RAG (sentence-transformers)

Serving & Cloud: FastAPI, Docker, GitHub Actions, Hugging Face Spaces, Azure, AWS, Cloudflare, Supabase, Firebase

Languages: Python, SQL, JavaScript/TypeScript, Dart, Rust, C++, C, Java, Kotlin

PROJECTS

DEPIGrad: Explainable Fraud Detection | *DEPI capstone* | *XGBoost, SHAP, Gemini, FastAPI, Svelte* | *github.com/Tu2525/DEPIGrad*

- XGBoost classifier on 310 IEEE-CIS-style transaction features with a tuned decision threshold (0.53): test ROC AUC 0.971, precision 0.886, recall 0.745, F1 0.809. At this threshold about a quarter of fraud is missed.
- SHAP picks the top 3 features per prediction and Gemini turns them into a short plain-English explanation, with a template fallback. Dockerized, live on Hugging Face Spaces.

Healthica: Smart Band Health Monitor | *Winner, YLF Hackathon 2026 (team of 5)* | *2nd place, Banha* | *Flutter, React, ESP32, TFLite*

- Trained and compared linear, random forest, gradient boosting and small neural nets on PhysioCGM (3,798 windows) and GlucoBench; exported 8 to 24 KB INT8 TFLite models that run on the phone.
- Honest results (random 80/20 split): sensor-only R^2 0.20, MAE 21.7 mg/dL. With one prior glucose reading R^2 is 0.97, but copying that reading alone gives 0.95, so the sensors add little. Forecast MAE 8.2, 11.4, 18.0 mg/dL at 15, 30, 60 minutes.

safi: Local Agentic LLM Backend | *in progress since Oct 2026* | *Python, llama.cpp, Qwen, sentence-transformers*

- Planner/executor design for a 16 GB laptop GPU (Qwen3.8-27B plans, Qwen3.6-35B-A3B executes). Built a model onboarding probe measuring VRAM, prefill/decode speed and thermals, plus a tool registry and CPU embedder with unit tests.

Atlas: Adaptive Tech Learning Assistant | *Flutter, FastAPI, Gemma 3*

- 1st place, Leaders of Tomorrow hackathon. AI mentor that predicts a learning track from chat, quizzes you and recommends resources; Gemma 3 (4B via Ollama), LangChain, FastAPI.
- Open source: merged fix in librespot (#1786, Rust Spotify library). Open PRs to Sonora (Spotify Connect, Jams) and MeasureIt_ARCH (#309).

EXPERIENCE

Generative AI Engineer, Digital Egypt Pioneers Initiative — *Cairo, Egypt (Hybrid) · Nov 2025 – Jul 2026*

- Built the DEPIGrad capstone end to end: model, FastAPI service, Svelte UI, Docker, Hugging Face Spaces (see Projects)

Backend & AI Engineer Intern, Aispeak — *Madrid, Spain (Remote) · Jul 2025 – Jan 2026*

- Built a backend pipeline routing user messages to Azure, processed by a LLaMA-70B model
- Stood up real-time messaging on Supabase; owned database schema design and API development

Backend Engineer Intern, Jiffy — *Remote · Aug 2025 – Dec 2025*

- Built core backend services: a multi-featured calendar and a containerized auth system in Docker

Instructor & Projects Contributor, Pixels Egypt Org. — *Helwan, Egypt · Oct 2023 – Oct 2025*

- Taught ML, Computer Vision and Flutter to 15+ students; 95% satisfaction, 80% project completion

Data Analyst Intern, National Telecommunication Institute — *Giza, Egypt · Dec 2024 – Mar 2025*

- Cleaned and structured raw telecom datasets in Python/Pandas for exploratory and inferential analysis

EDUCATION & CERTIFICATIONS

B.Sc., Intelligent Systems Engineering, Helwan National University — *Helwan, Egypt · 2022 – 2027 (expected)*

DEPI Generative AI and Mobile tracks | Stanford ML Specialization | Huawei HCIP-AI | AWS Cloud Practitioner | NLP (NTI)